

A Practical Guide to Regularization machine learning

TOPIC -- REGULARIZATION MACHINE LEARNING

-- 01 --

If W_{ij} encodes the result of some distance metric for points x_i and x_j , it is desirable that $f(x_i) \approx f(x_j)$. This regularizer captures this intuition, and is equivalent to:

-- 02 --

Group sparsity without overlaps Groups of features can be regularized by a sparsity constraint, which can be useful for expressing certain prior knowledge into an optimization problem. In the case of a linear model with non-overlapping known groups, a regularizer can be defined:

-- 03 --

$R(w) = \sum \sigma(W) \geq 1$ where $\sigma(W)$ is the eigenvalues in the singular value decomposition of W .

-- 04 --

$$L_S[f_n] = \frac{1}{n} \sum_{i=1}^N V(f_n(\hat{x}_i), \hat{y}_i)$$

-- 05 --

Early stopping can be viewed as regularization in time. Intuitively, a training procedure such as gradient descent tends to learn more and more complex functions with increasing iterations. By regularizing for time, model complexity can be controlled, improving generalization. Early stopping is implemented using one data set for training, one statistically independent data set for validation and another for testing. The model is trained until performance on the validation set no longer improves and then applied to the test set.

-- 06 --

where X and Y are the domains of input data x and their labels y respectively. Typically in learning problems, only a subset of input data and labels are available,

measured with some noise. Therefore, the expected error is unmeasurable, and the best surrogate available is the empirical error over the N available samples:

-- 07 --

$$\text{prox}_{\lambda, R, g}(\mathbf{w}_g) = \begin{cases} (1 - \lambda \|\mathbf{w}_g\|_2) \mathbf{w}_g, & \text{if } \|\mathbf{w}_g\|_2 > \lambda \\ 0, & \text{if } \|\mathbf{w}_g\|_2 \leq \lambda \end{cases}$$

$$\text{prox}_{\lambda, R, g}(\mathbf{w}_g) = \begin{cases} \left(1 - \frac{\lambda}{\|\mathbf{w}_g\|_2}\right) \mathbf{w}_g, & \text{if } \|\mathbf{w}_g\|_2 > \lambda \\ 0, & \text{if } \|\mathbf{w}_g\|_2 \leq \lambda \end{cases}$$

-- 08 --

In mathematics, statistics, finance, and computer science, particularly in machine learning and inverse problems, regularization is a process that converts the answer to a problem to a simpler one. It is often used in solving ill-posed problems or to prevent overfitting. Although regularization procedures can be divided in many ways, the following delineation is particularly helpful:

-- 09 --

The exact solution to the unregularized least squares learning problem minimizes the empirical error, but may fail. By limiting T , the only free parameter in the algorithm above, the problem is regularized for time, which may improve its generalization. The algorithm above is equivalent to restricting the number of gradient descent iterations for the empirical risk

-- 10 --

where $(\hat{x}_i, \hat{y}_i)$, $1 \leq i \leq n$, would represent samples used for training. In the case of a general function, the norm of the function in its reproducing kernel Hilbert space is:

-- 11 --

Regularization in machine learning In machine learning, a key challenge is enabling models to accurately predict outcomes on unseen data, not just on familiar training data. Regularization is crucial for addressing overfitting—where a model memorizes training data details but cannot generalize to new data. The goal of regularization is to encourage models to learn the broader patterns within the data rather than memorizing it. Techniques like early stopping, L1 and L2 regularization, and dropout are designed to prevent overfitting and underfitting, thereby enhancing the model's ability to adapt to and perform well with new data, thus improving model generalization.

-- 12 --

These techniques are named for Andrey Nikolayevich Tikhonov, who applied regularization to integral equations and made important contributions in many other areas. When learning a linear function f , characterized by an unknown vector w such that $f(x) = w \cdot x$, one can add the L_2 -norm of the vector w to the loss expression in order to prefer solutions with smaller norms. Tikhonov regularization is one of the most common forms. It is also known as ridge regression. It is expressed as:

-- 13 --

$$\text{prox}_{R^D}(v) = \operatorname{argmin}_{w \in R^D} \{ R(w) + \frac{1}{2} \|w - v\|^2 \},$$

-- 14 --

$$I[f_n] = \int_{X \times Y} V(f_n(x), y) \rho(x, y) dx dy$$

-- 15 --

Regularizers for multitask learning In the case of multitask learning, T problems are considered simultaneously, each related in some way. The goal is to learn T functions, ideally borrowing strength from the relatedness of tasks, that have predictive power. This is equivalent to learning the matrix $W : T \times D$.

-- 16 --

$$\min_w \sum_{i=1}^n V(\hat{x}_i \cdot w, \hat{y}_i) + \lambda \|w\|^2,$$

-- 17 --

$R(f_1 \cdots f_T) = \sum_{r=1}^C \sum_{t \in I(r)} \|f_t - \frac{1}{|I(r)|} \sum_{s \in I(r)} f_s\|_{H_k}^2$ where $I(r)$ is a cluster of tasks. This regularizer is similar to the mean-constrained regularizer, but instead enforces similarity between tasks within the same cluster. This can capture more complex prior information. This technique has been used to predict Netflix recommendations. A cluster would correspond to a group of people who share similar preferences.